

A mass extinction known as the **Great Dying** wiped out 90 percent of all species on Earth about **252 million years ago**.

How new species evolve

A species is a group of organisms that look the same and survive in the same way, and can breed together to produce viable young in the wild. Some species of bat look very similar to one another and live in the same areas, but attract mates using different calls and cannot breed with each other. Speciation is the formation of a new species. Species can evolve sympatrically (from one ancestor) or allopatrically (when populations are isolated).

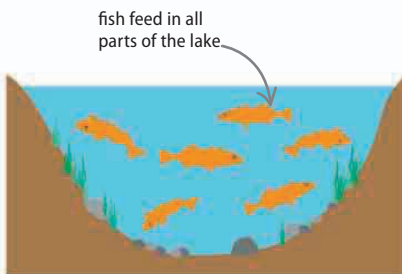
REAL WORLD

Extinction



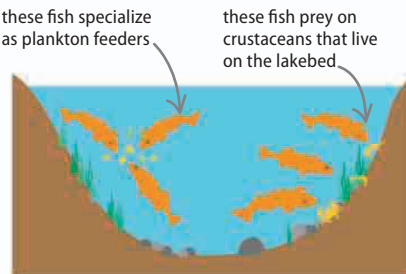
Most of the knowledge of evolution comes from fossils of extinct species. An extinct species is one that has no members left alive. Fossils, such as this primitive bird, form when body parts are replaced with rocky minerals over a long period of time. Fossils show us what the ancestors of today's species looked like. If the lost species died out after it evolved into another species, experts call this pseudoextinction.

sympatric speciation



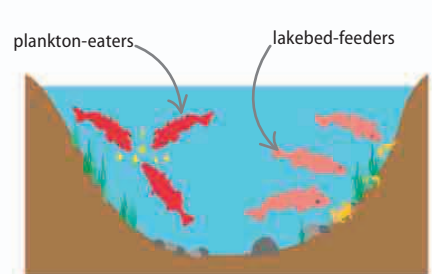
△ 1. One species of fish

One species of fish lives in the lake. The fish feed mainly on small animals in the water and on the lakebed.



△ 2. Specialist feeders

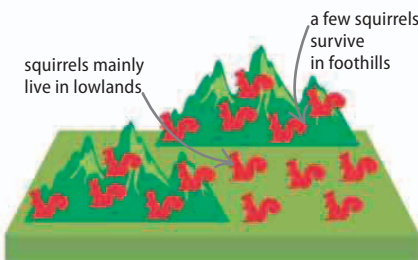
Gradually, the fish split into two groups as they specialize on catching certain animals. The groups evolve in different ways to exploit the different food sources.



△ 3. Two species

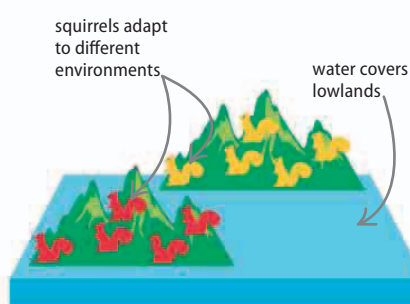
The groups rarely mix, and eventually they evolve into two distinct species that cannot breed with each other any more.

allopatric speciation



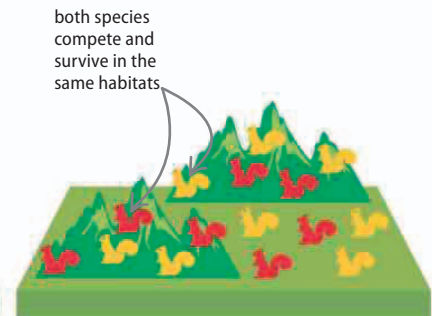
△ 1. One species of squirrel

A species of squirrel has spread across a wide range. Although they do not meet, squirrels from either end of the range could breed with one another.



△ 2. Geographic isolation

Rising sea levels turns the range into isolated islands. Squirrels on both islands adapt to life in their own mountain habitat, forming two new species.



△ 3. Sharing habitat

The sea level falls again, creating a new lowland habitat for the squirrels to exploit. Both species mix throughout the original range, but are no longer able to breed.